ಬಿ.ಎಂ.ಎಸ್. ತಾಂತ್ರಿಕ ಮತ್ತು ವ್ಯವಸ್ಥಾಪನಾ ಮಹಾವಿದ್ಯಾಲಯ
**BMS Institute of Technology and Management**
(An Autonomous Institution Affiliated to VTU, Belagavi)
Avalahalli, Doddaballapur Main Road, Bengaluru – 560064

# Programme: B E – Computer Science and Engineering/ Information Science Engineering

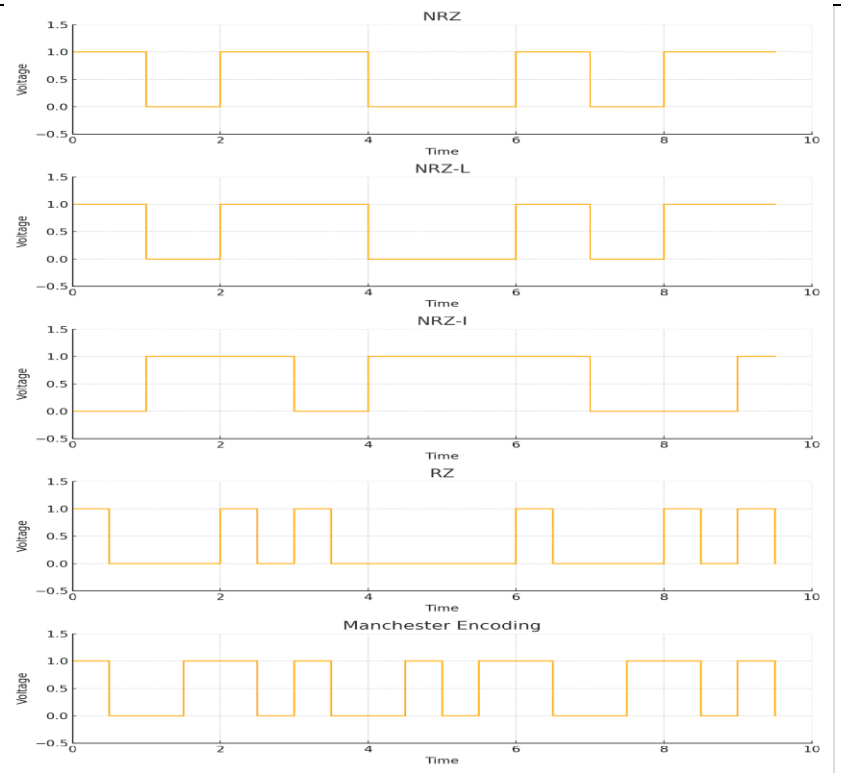
## Scheme - Internal Assessment – I

| TERM: | | COURSE NAME: COMPUTER NETWORKS |
|---|---|---|
| DATE: 25/11/2024 TIME: 2 to 3:30 pm | | COURSE CODE: BCS502 |
| SEMESTER: V | MAX MARKS: 40 | COURSE COORDINATOR: Dr. Anil G N, Dr. Swetha M S, Dr. Gireesh Babu, Dr. Kalai Vani Y.S , Dr. Goutami , Mr. Akshay |

Answer **Four** Full Questions **Max. Marks: 40**

| Q. NO | Questions | Blooms Levels (L1 to L6) | CO | Marks |
|---|---|---|---|---|
| 1.a | You work in an office where employees face slow file transfers and connectivity issues. The IT team has set up a data communication system to enable smooth communication and data sharing.<br>1) Identify the importance of Data Communication in the office scenario?<br>2) Select key characteristics of data communication that could be affecting the system's performance.<br>**SOLUTION:**<br>**1)Importance of Data Communication**<br>• **D**ata communication systems enable faster and more reliable sharing of files, minimizing delays caused by slow transfers. This improves productivity by ensuring employees spend less time waiting for data to load or transfer.<br>• A robust data communication system ensures stable and uninterrupted connectivity among devices and team members. This is essential for smooth collaboration, especially in hybrid work environments where remote and on-site staff need to stay connected.<br>• Modern data communication systems incorporate encryption and secure protocols to protect sensitive office data during transmission. This reduces the risk of data breaches and cyberattacks.<br>**2) Key characteristics of data communication could be affecting the system's performance.** | L3 | CO1 | |

| | | | | |
|---|---|---|---|---|
| | ***Telecommunication****-communication at a distance.*<br>***Data*** *-information presented in whatever form is agreed upon by the parties creating and using the data.*<br>***Data communications*** *are the exchange of data between two devices via some form of transmission medium such as a wire cable.*<br>***Fundamental characteristics[Effectiveness of DC]:***<br>1. ***Delivery****-correct destination*<br>2. ***Accuracy****-deliver data accurately*<br>3. ***Timeliness-*** *delivery data in timely manner*<br>4. ***Jitter****-variation in packet arrival time.*<br><br>**(i)+(ii) =3m+ 3m** | | | **6** |
| **b** | A software development company is designing a new chat application that will allow users to send text messages, images, and voice notes. The application must ensure data integrity and efficient data transfer over the network. The developers are using a protocol layering approach to design the application's network functionality.<br><br>Organize how protocol layering will benefit the development of this chat application in terms of design flexibility and troubleshooting.<br><br>**1) Design Flexibility**<br>**2) Trouble shooting**<br>**3) Scalability**<br>**4) Reusability**<br><br>**Each point (1m+1m+1m+1m)=4m** | **L3** | **CO1** | **4** |
| **OR** | | | | |
| **2.a** | Suppose we are sending data having sequence 1011001011 from device A to B. Apply following schemes: NRZ, NRZ-L, NRZ-I, RZ and Manchester scheme for given sequence. | **L3** | **CO1** | |



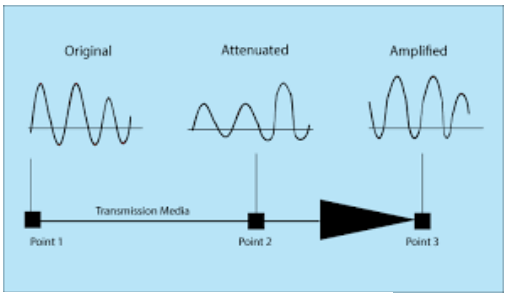

**NRZ(1m)+NRZ-L(1m)+NRZ-I(1m)+RZ(1.5m)+Manchester scheme(1.5M) = 6m**

**b** A retail company has set up a Wide Area Network (WAN) to connect its headquarters in New York to its warehouses in remote locations across the country. However, during data transfers, the IT team notices that files often get corrupted, video calls are choppy, and download speeds vary significantly.
After conducting an initial analysis, the team identifies three potential causes of transmission impairments: **attenuation**, **noise**, and **delay distortion**.

Based on the scenario, identify how **attenuation**, **noise**, and **delay distortion** might be contributing to the network issues.
SOLUTION:

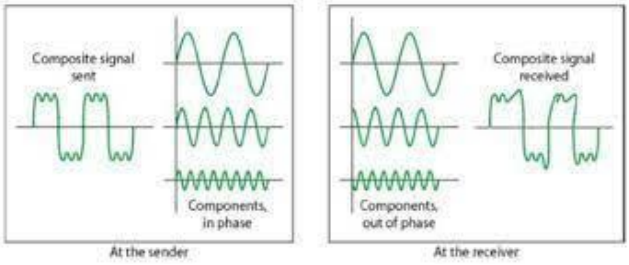
**L3** **CO1** **4**

1. **Attenuation-2M**

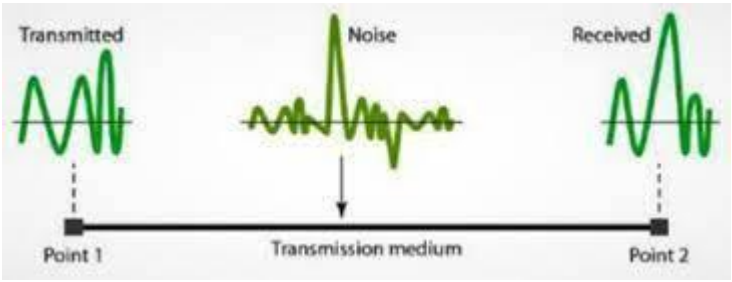


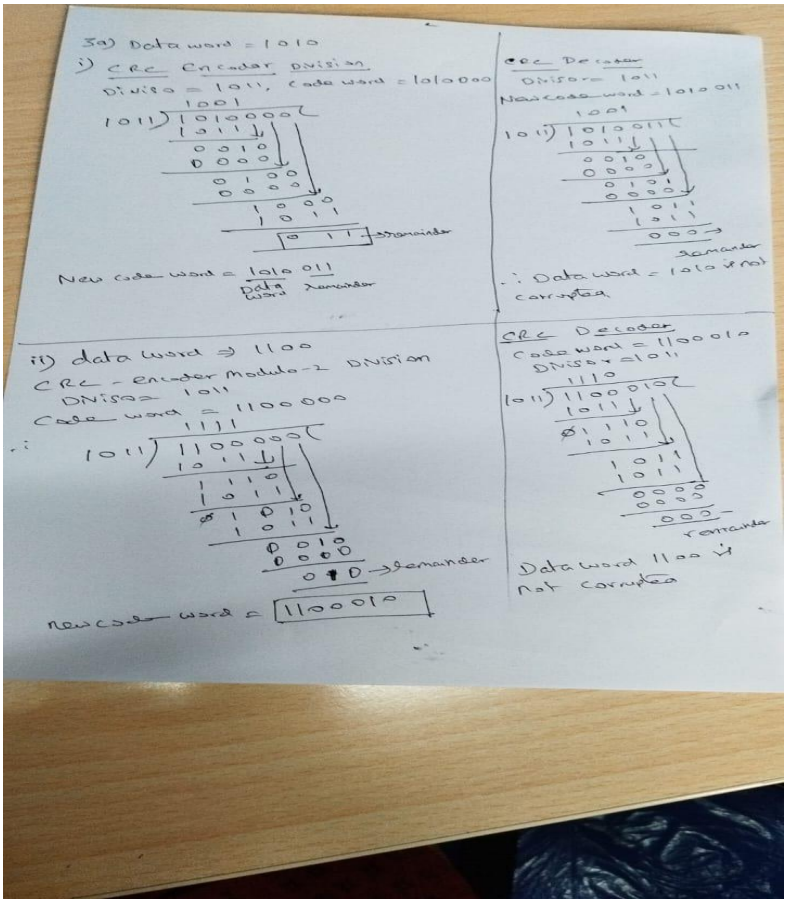

$$dB = 10 \log_{10} \frac{P_2}{P_1}$$

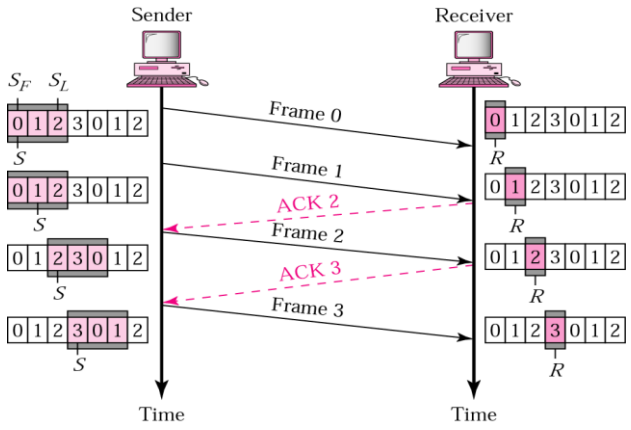
2. **Distortion-2M**

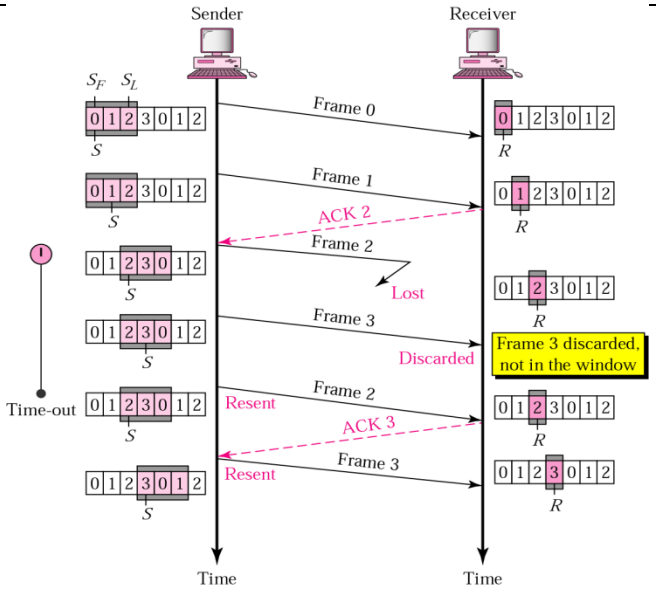



**3. Noise-2M**

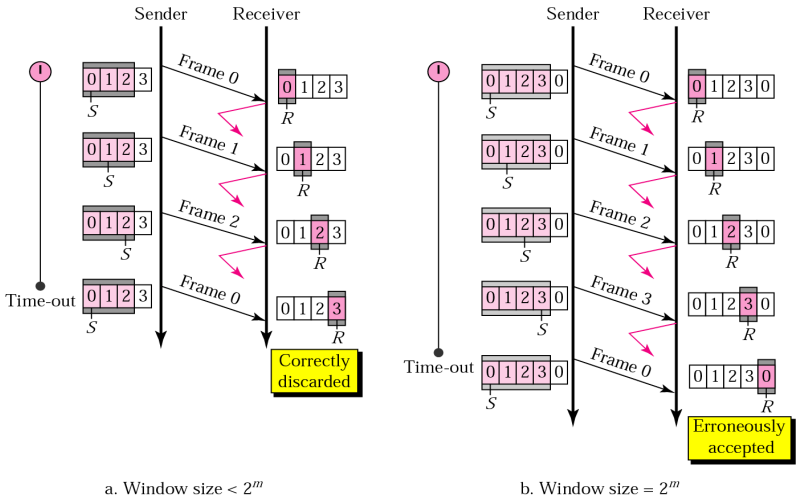



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| **3.a** | Examine three code bits added to 1010 data words in CRC encoding if divisor is 1011? Show complete division. And also, on the decoder or receiver side, show how to check if data word 1100 is corrupted or not with complete division.<br><br>  <br><br>3m + 3m=6m | **L4** | **CO1** | **6** |

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| **b** | An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. Solve how many data elements are carried by each signal element and how many signal elements do we need?<br>**Solution:**<br>Data element per signal is given as:<br>Data element(N)= Bit Rate/Baud Rate<br>=8000/1000<br>N = 8 bits<br>Signal elements (S) = $2^N$<br>where N is data elements(N=8)<br>$S=2^8$ =256<br>**Calculation of data element(2m)+signal elements(2m)=4m** | **L3** | **CO1** | **4** |
| | **OR** | | | |
| **4.a** | Analyze the working Go-Back-N with the following scenario with proper flow and timeline diagram.<br>i. When all frames are reached successfully<br>ii. When certain frames are lost<br>iii. When acknowledgement is lost.<br>**i. When all frames are reached successfully-2M**<br>**ii. When certain frames are lost-2M**<br>**iii. When acknowledgement is lost--2M**<br>**i. When all frames are reached successfully**<br>  <br><br>**ii. When certain frames are lost** | **L4** | **CO1** | **6** |




### iii. When acknowledgement is lost




a. Window size < $2^m$

b. Window size = $2^m$

| | | | | |
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| **b** | A voice grade channel of a telephone network has a bandwidth of 3.4 kHz. Solve the following based on given condition.<br>(a) Channel capacity for S/N= 30dB<br>(b) S/N required supporting information transfer at 4800 bps. | **L3** | **CO1** | **4** |

6.a) i, Given that

Bandwidth (B) = 3.4 KHz and

S/N = 30dB

$30 = \log_{10} SNR$

SNR = Antilog 30

SNR = 1000

Information capacity of telephone channel is given by:

$C = B \log_2 (1+SNR)$

$C = 3.4 \times 10^3 \log_2 (1+1000)$

$C = 3.4 \times 10^3 \log_2 (1001)$

$C = 3400 \times 9.9672$

C = 33888.57 bits per second (bps)

**C = 33.89 Kbps**

6.a) ii, Given, Bandwidth = 3.4 KHz

Capacity (C) = 4800 bps

$\therefore C = B \times \log_2 (1+SNR)$

$4800 = 3.4 \times 10^3 \times \log_2 (1+SNR)$

$\frac{4800}{3.4 \times 10^3} = \log_2 (1+SNR)$

$\frac{4800}{3.4} \times 10^{-3} = \log_2 (1+SNR)$

$1.4117 = \log_2 (1+SNR)$

$1+SNR = 2^{1.4117}$

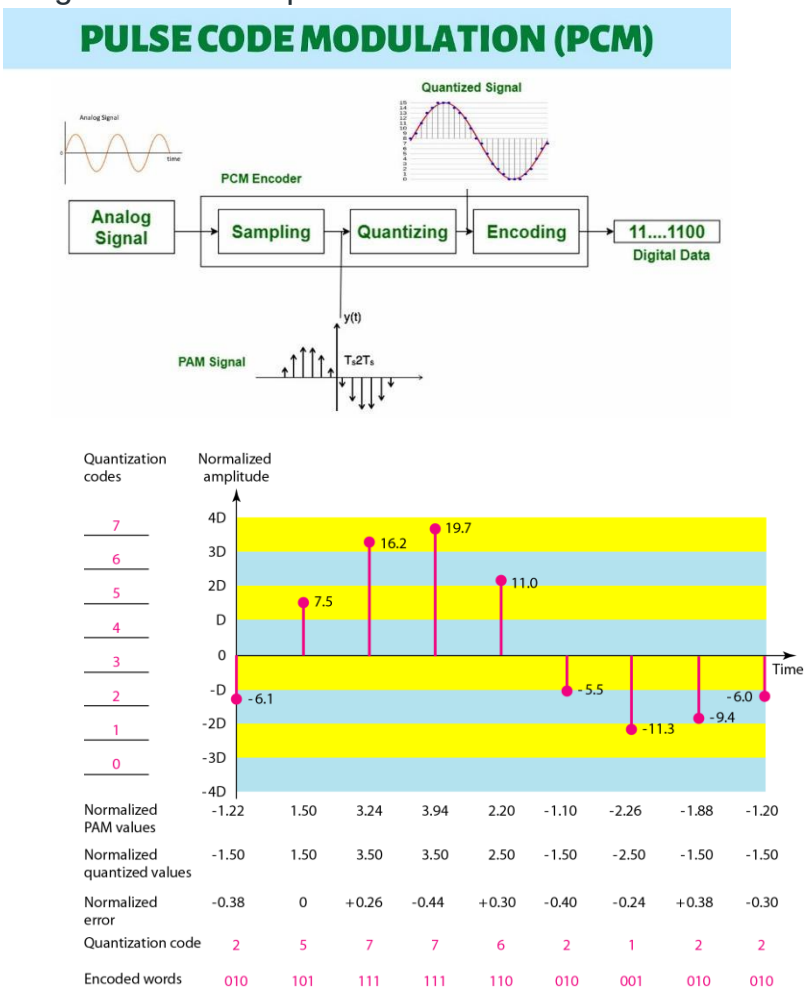
$1+SNR = 2.6605$

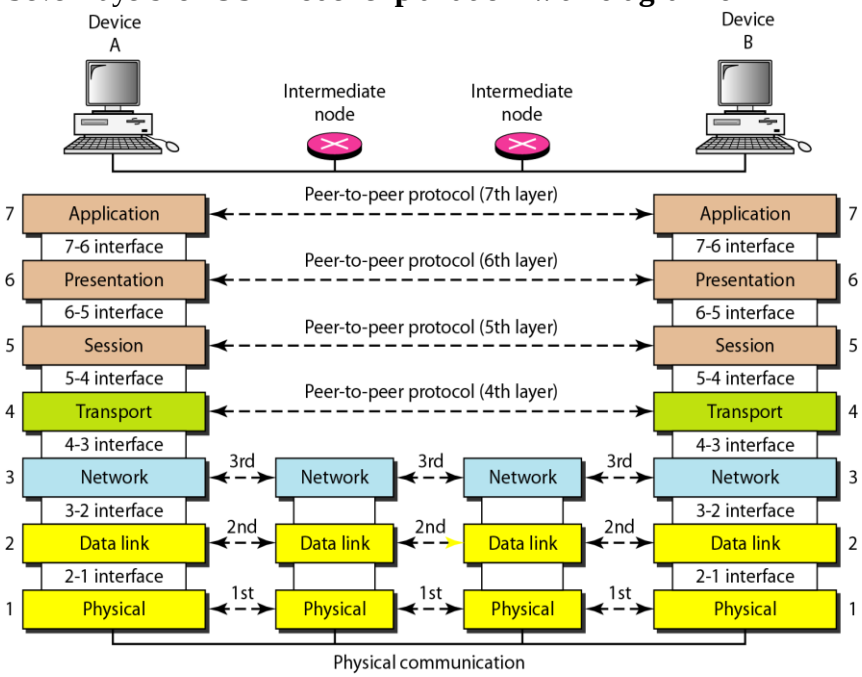
$SNR = 2.6605 - 1$

**SNR = 1.6605**

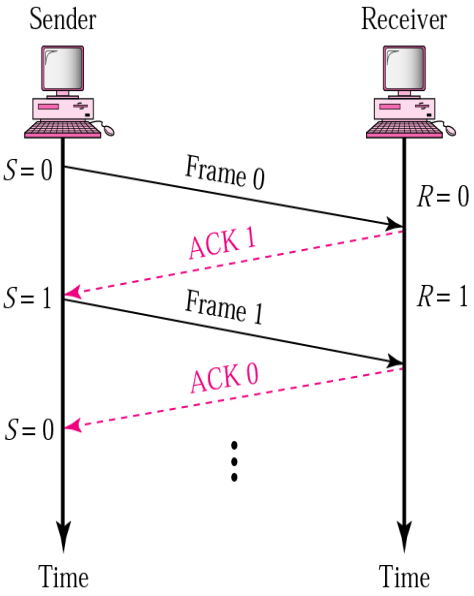
$\therefore S/N = 1.6605$

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| **5.a** | Analyze why we need Quantization process in PCM encoder with an example.<br>A PCM encoder has the following three processes | **L4** | **CO2** | **6** |

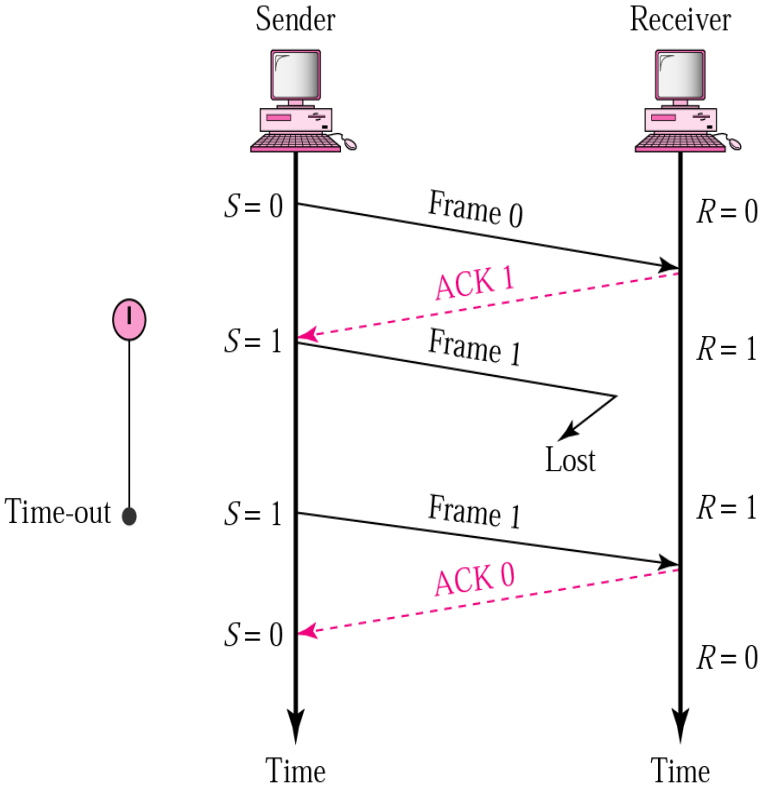
| | | | | |
|---|---|---|---|---|
| | Diagram-2M+3 steps 4M<br>  <br> | | | |
| **b** | A large university campus with multiple departments and buildings wants to set up a robust network that can handle heavy data traffic while ensuring high reliability and minimal downtime. The network should be scalable to accommodate new buildings and departments as the campus expands. Additionally, quick fault identification and isolation are crucial to minimize disruptions during academic sessions. The university's IT team is considering different network topologies: **Star**, **Mesh**, **Bus**, and **Ring**.<br><br>Select which network topology would you recommend for the university's campus network and identify limitations or potential challenges with implementing this topology in a campus environment.<br><br>**Preferred Topology is: Star or Mesh Topology**<br><br>• **Explaining all 4 topologies with diagram-4M**<br><br>**Star**, **Mesh**, **Bus**, and **Ring** | **L3** | **CO2** | **4** |
| | **OR** | | | |
| **6.a** | You are part of an IT team in a company where employees use a local network to access shared files, send emails, and communicate via video calls. Recently, some users have faced issues with slow file transfers, failed email deliveries, and poor video call quality. To | **L4** | **CO2** | **6** |

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| | troubleshoot the problem, your team decides to analyze the situation using the OSI (Open Systems Interconnection) model, which divides network communication into seven layers.<br><br>Compare the seven layers of the OSI model, and identify the cause of issues like slow file transfers or failed email deliveries in the given scenario?<br><br>Seven layers of OSI-model **explanation with diagram-6M**<br>  <br> | | | |

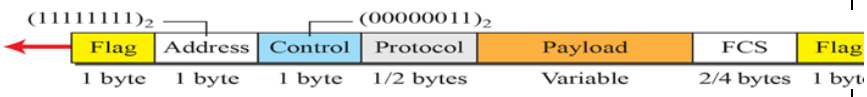
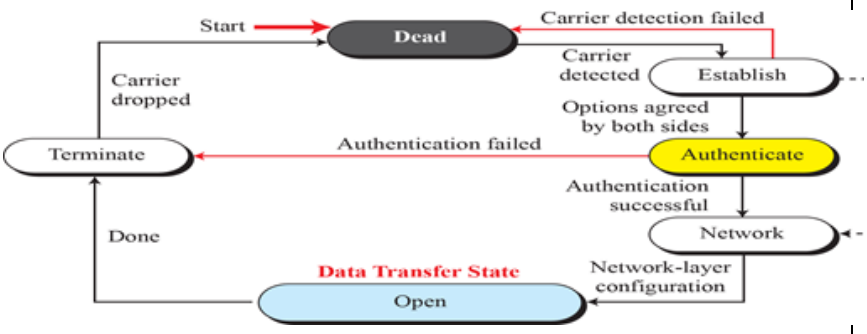
<table>
<tr>
<td>b</td>
<td>In a busy office network, employees are experiencing frequent data collisions while sharing files. The IT team is considering using ALOHA protocols to improve network performance. Choose main difference between Pure ALOHA and Slotted ALOHA.<br><br>
<table>
<tr><th>Pure Aloha</th><th>Slotted Aloha</th></tr>
<tr><td>In this Aloha, any station can transmit the data at any time.</td><td>In this, any station can transmit the data at the beginning of any time slot.</td></tr>
<tr><td>In this, The time is continuous and not globally synchronized.</td><td>In this, The time is discrete and globally synchronized.</td></tr>
<tr><td>Vulnerable time for Pure Aloha = 2 x Tt</td><td>Vulnerable time for Slotted Aloha = Tt</td></tr>
<tr><td>In Pure Aloha, the Probability of successful transmission of the data packet<br>= G x e-2G</td><td>In Slotted Aloha, the Probability of successful transmission of the data packet<br>= G x e-G</td></tr>
<tr><td>In Pure Aloha, Maximum efficiency<br>= 18.4%</td><td>In Slotted Aloha, Maximum efficiency<br>= 36.8%</td></tr>
<tr><td>Pure Aloha doesn’t reduce the number of collisions to half.</td><td>Slotted Aloha reduces the number of collisions to half and doubles the efficiency of Pure Aloha.</td></tr>
</table>
<br>Any three differences with diagrams.</td>
<td>L3</td>
<td>CO2</td>
<td>4</td>
</tr>
<tr>
<td>7.a</td>
<td>You are managing a network for a small company, where employees regularly send data packets to a central server. However, you’ve noticed occasional delays or data loss when sending large files. After investigating, you realize that the network may be experiencing packet loss due to unreliable communication. The company decides to implement Stop-and-Wait ARQ (Automatic Repeat Request) to ensure reliable transmission of data.<br><br>Examine how Stop-and-Wait ARQ ensures reliable data transmission, handle packet loss or errors.<br><br>Reliable Data Transmission:</td>
<td>L4</td>
<td>CO2</td>
<td>6</td>
</tr>
</table>




**Handling packet loss or errors:**




| | | | | |
|---|---|---|---|---|
| **b** | Identify the need of classless addressing.<br><br>**In classful addressing, a large part of the available addresses were wasted.**<br><br>Find the class of the following classful IP addresses:<br><br>i)201.20.31.65-**Class**: **C**<br><br>ii)126.11.11.239-**Class**: **A** | **L3** | **CO2** | **4** |

| | | | | |
|---|---|---|---|---|
| | iii)193.14.56.22- **Class: C**<br><br>iv) 130.34.54.12-**Class: B** | | | |
| | **OR** | | | |
| **8.a** | Traditional modems are connected by Point to Point Protocol at the data link layer. Analyze how PPP manage the connections.<br><br>· Point to Point Protocol format-2M<br>  <br><br>· Transition phases-4M<br>  <br> | **L4** | **CO2** | **6** |
| **b** | Stations in a pure Aloha network send frames of size 1000 bits at the rate of 1 Mbps. Identify vulnerable time for this network? Repeat it for Slotted Aloha also.<br><br>**Answer:**<br><br>For a pure ALOHA network, the vulnerable time is equal to the time required for a frame to be transmitted by a station, plus the time required for the same frame to be transmitted by another station (assuming the maximum possible distance between the stations).<br><br>To calculate the vulnerable time, we need to first calculate the time required for a station to transmit a 1000-bit frame at a rate of 1 Mbps:<br><br>Time to transmit one frame = (frame size) / (transmission rate)<br><br>= 1000 / 1,000,000<br><br>= 0.001 seconds<br><br>Next, we need to calculate the time required for the same frame to be transmitted by another station, assuming the maximum possible distance between the stations. We can use the speed of light to calculate the time required for the signal to travel the maximum | **L3** | **CO2** | **4** |

| | | | | |
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| | distance:<br><br>Time for signal to travel maximum distance = (maximum distance) / (propagation speed)<br><br>= (2 * maximum distance) / (propagation speed) // Because the signal has to travel to and from the station<br><br>= (2 * 2 * 10^4) / (3 * 10^8) // Assuming a distance of 20 km and a propagation speed of 3 x 10^8 m/s = 0.000133 seconds<br><br>Therefore, the vulnerable time for the pure ALOHA network is the sum of the time required for a station to transmit a frame and the time required for the same frame to be transmitted by another station:<br><br>Vulnerable time = time to transmit one frame + time for signal to travel maximum distance = 0.001 + 0.000133 = 0.001133 seconds<br><br>So, the vulnerable time for this pure ALOHA network is 0.001133 seconds. This means that a station's transmission can collide with another transmission during this duration. | | | |